

Операционные системы

Продвинутое использование git

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы



Получение навыков правильной работы с репозиториями git.



- 1 Выполнить работу для тестового репозитория.
- 2 Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

2. Процесс выполнения лабораторной работы



```
irabzalimov@irabzalimov:~$ pnpm setup
Appended new lines to /home/irabzalimov/.bashrc

Next configuration changes were made:
export PNPM_HOME="/home/irabzalimov/.local/share/pnpm"
case ":$PATH:" in
  *:$PNPM_HOME:*) ;;
  *) export PATH="$PNPM_HOME:$PATH" ;;
esac

To start using pnpm, run:
source /home/irabzalimov/.bashrc
irabzalimov@irabzalimov:~$ source ~/.bashrc
irabzalimov@irabzalimov:~$
```

Рисунок 1: Node.js

```
irabzalimov@irabzalimov:~$ pnpm add -g commitizen
```

```
Update available! 10.33.0 → 12.8.1.  
Changelog: https://pnpm.io/v/12.8.1  
To update, run: pnpm self-update
```

```
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
```

```
Packages: +148
```

```
Downloading @typescript/typescript-linux-x64@7.0.2: 9.67 MB/9.67 MB, done
```

```
Progress: resolved 167, reused 0, downloaded 148, added 148, done
```

```
/home/irabzalimov/.local/share/pnpm/global/5:
```

```
+ commitizen 4.3.2
```

```
Done in 3s using pnpm v10.33.0
```

```
irabzalimov@irabzalimov:~$
```

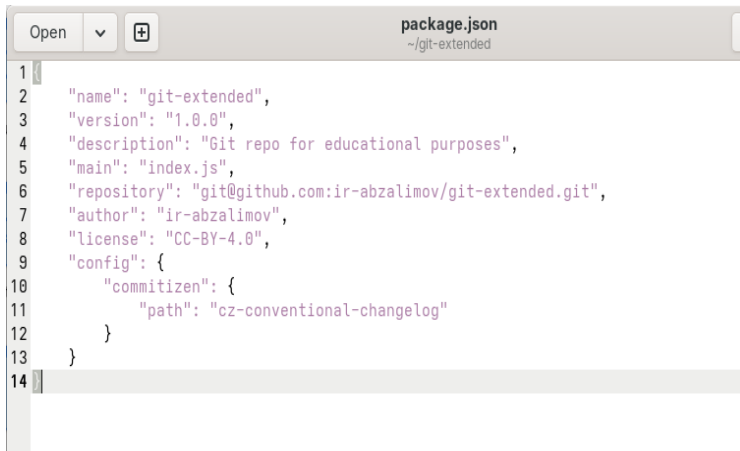
Рисунок 2: установка commitizen

```
irabzalimov@irabzalimov:~$ pnpm add -g standard-changelog
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +14
++++
Progress: resolved 181, reused 148, downloaded 14, added 14, done

/home/irabzalimov/.local/share/pnpm/global/5:
+ standard-changelog 8.1.0

Done in 2.1s using pnpm v10.33.0
irabzalimov@irabzalimov:~$
```

Рисунок 3: установка standard-changelog



The image shows a code editor window with the title bar "package.json" and the path "~/git-extended". The editor contains the following JSON content:

```
1 {  
2   "name": "git-extended",  
3   "version": "1.0.0",  
4   "description": "Git repo for educational purposes",  
5   "main": "index.js",  
6   "repository": "git@github.com:ir-abzalimov/git-extended.git",  
7   "author": "ir-abzalimov",  
8   "license": "CC-BY-4.0",  
9   "config": {  
10     "commitizen": {  
11       "path": "cz-conventional-changelog"  
12     }  
13   }  
14 }
```

Рисунок 4: package.json



```
irabzalimov@irabzalimov:~/git-extended$ gedit package.json
irabzalimov@irabzalimov:~/git-extended$ git add .
irabzalimov@irabzalimov:~/git-extended$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:    Other changes that don't
modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 44a8d87] chore: test
 1 file changed, 14 insertions(+)
 create mode 100644 package.json
irabzalimov@irabzalimov:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 483 bytes | 483.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ir-abzalimov/git-extended.git
 5808185..44a8d87  master -> master
irabzalimov@irabzalimov:~/git-extended$
```

Рисунок 5: Отправка

```
irabzalimov@irabzalimov:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? []
Hooks and filters directory? [/home/irabzalimov/git-extended/.git/hooks]
irabzalimov@irabzalimov:~/git-extended$ git branch
* develop
  master
irabzalimov@irabzalimov:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/ir-abzalimov/git-extended/pull/new/develop
remote:
To github.com:ir-abzalimov/git-extended.git
 * [new branch]      develop -> develop
irabzalimov@irabzalimov:~/git-extended$
```

Рисунок 6: Инициализация и загрузка изменений



```
irabzalimov@irabzalimov:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
Merge made by the 'ort' strategy.
  CHANGELOG.md | 1 +
  1 file changed, 1 insertion(+)
  create mode 100644 CHANGELOG.md
Already on 'master'
Your branch is ahead of 'origin/master' by 2 commits.
  (use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
  CHANGELOG.md | 1 +
  1 file changed, 1 insertion(+)
  create mode 100644 CHANGELOG.md
Deleted branch release/1.0.0 (was 9b67641).

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged '1.0.0'
- Release tag '1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

irabzalimov@irabzalimov:~/git-extended$
```

Рисунок 7: Завершение релиза

```
irabzalimov@irabzalimov:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 550 bytes | 550.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To github.com:ir-abzalimov/git-extended.git
  44a8d87..d517536  develop -> develop
  44a8d87..d4a0e75  master -> master
irabzalimov@irabzalimov:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 157 bytes | 157.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ir-abzalimov/git-extended.git
 * [new tag]         1.0.0 -> 1.0.0
irabzalimov@irabzalimov:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/ir-abzalimov/git-extended/releases/tag/v1.0.0
irabzalimov@irabzalimov:~/git-extended$
```

Рисунок 8: Отправка



```
irabzalimov@irabzalimov:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing to your feature. When done, use:

    git flow feature finish feature_branch

irabzalimov@irabzalimov:~/git-extended$ touch test2
irabzalimov@irabzalimov:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was d517536).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

irabzalimov@irabzalimov:~/git-extended$
```

Рисунок 9: Объединение веток

```
Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged '1.2.3'
- Release tag '1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

irabzalimov@irabzalimov:~/git-extended$ ^[[200~git push --all
bash: git: command not found...
irabzalimov@irabzalimov:~/git-extended$ ~git push --all
bash: ~git: command not found...
Similar command is: 'git'
irabzalimov@irabzalimov:~/git-extended$ git push --all
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 592 bytes | 592.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
To github.com:ir-abzalimov/git-extended.git
   d517536..1b0f9e5  develop -> develop
   d4a0e75..5bf75f6  master -> master
irabzalimov@irabzalimov:~/git-extended$ git push --all
Everything up-to-date
irabzalimov@irabzalimov:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/ir-abzalimov/git-extended/releases/tag/v1.2.3
irabzalimov@irabzalimov:~/git-extended$
```

Рисунок 10: Завершение релиза



```
git add .
irabzalimov@irabzalimov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:    Other changes that don't
modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 6c204b4] chore: test
 1 file changed, 11 insertions(+), 11 deletions(-)
irabzalimov@irabzalimov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 478 bytes | 478.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ir-abzalimov/2026-1--study--os-intro.git
   e10c6e3..6c204b4  master -> master
irabzalimov@irabzalimov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 11: package.json и коммит



```
Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged '1.0.0'
- Release tag '1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

irabzalimov@irabzalimov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 647 bytes | 647.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To github.com:ir-abzalimov/2026-1--study--os-intro.git
   6c204b4..11c1236  develop -> develop
   6c204b4..55334eb  master -> master

irabzalimov@irabzalimov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 156 bytes | 156.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:ir-abzalimov/2026-1--study--os-intro.git
   * [new tag]           1.0.0 -> 1.0.0

irabzalimov@irabzalimov:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 12: Завершение релиза

3. Выводы по проделанной работе



Мы приобрели практические навыки взаимодействия с дополнительными функциями гитхаб.